21ASC305T
ROCKET PROPULSION

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- ***Unit-1 - Basic Concepts 9 Hour***
- *Rocket propulsion systems -working principle, Classification of rockets, Rocket equation, Rocket Nozzles and its classifications -Nozzle performance- equilibrium and frozen flow, Characteristic velocity and Thrust coefficient, Performance parameters and efficiencies of rocket, Staging and clustering of rocket*
- ***Unit-2 - Chemical Propellants 9 Hour***
- *Chemical propellants -Molecular mass, Specific heat capacities, Specific heat ratio, mixture ratio, Stoichiometric ratio and equivalence ratio, Energy release during combustion-Heat of formation and Heat of combustion-Combustion Instabilities-Criterion for choice of propellant, Solid Propellants-Composition and processing, Liquid Propellants-Classifications of liquid propellant, Storability of liquid propellant, Chemical Equilibrium*
- ***Unit-3 - Solid Rocket Engine 9 Hour***
- *Propulsion Elements for Solid Rocket Motors, Solid propellant combustion and Internal Ballistics of Motors - Mechanism of burning, Ignition System-Igniter types - Solid propellant grain design - Burn Rate-Factors influencing burn rates, burn rate index for stable operation, Action time and burn time, Design of Solid Propellant rocket, Simulation of solid rocket motor performance.*
- ***Unit-4 - Liquid Rocket Engine 9 Hour***
- *Liquid Propellant Rocket-Hardware components and its functions, Propellant feed systems –Gas pressure feed system, Turbo pump feed systems and engine cycles, Tank Pressurization - Types of injectors – Thrust chamber, Cooling of Thrust chamber - Cryogenic propulsion system and its special features, Simulation of Liquid rocket performance, Hybrid rockets- Review of solid-fuel regression rate behavior in classical and non-classical Hybrid Rocket Motors - Rocket engine design using software.*
- ***Unit-5 - Non-conventional Propulsion Techniques 9 Hour***
- *Non - conventional propulsion techniques -Electrical Rockets-Electro-thermal propulsion system-Arc-jet thruster, Resistojet Thruster-Electro-static thruster-Ion thruster, Bombardment Ionization, Hall Effect Thruster -Electro-magnetic propulsion system-Magneto plasma dynamic accelerator, Pulsed Plasma accelerator- Laser Propulsion -Nuclear powered rockets- Solar Propulsion system*

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- *1. George P. Sutton, Oscar Biblarz, "Rocket propulsion elements", Wiley India Pvt Ltd. Ninth Edition, 2017*
- *2. Ramamurthi.K, "Rocket propulsion", Laxmi Publications, India, Second edition, 2019.*
- *3. J D. Mattingly and K M. Boyer,"Elements of Propulsion: Gas Turbines and Rockets", Second Edition,2016*
- *4. Philip Hill and Carl Peterson, "Mechanics and thermodynamics of propulsion", Pearson India, second edition, 2010.*
- *5. Stephen R. Turns, "An Introduction to Combustion: Concepts", McGraw-Hill Education; 4th edition, 2020*
- *6. Yahya S M, "Gas Tables",8th Edition, 2018*

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Unit-3 - Solid Rocket Engine

- *Propulsion Elements for Solid Rocket Motors, Solid propellant combustion and Internal Ballistics of Motors -Mechanism of burning, Ignition System-Igniter types - Solid propellant grain design - Burn Rate-Factors influencing burn rates, burn rate index for stable operation, Action time and burn time, Design of Solid Propellant rocket, Simulation of solid rocket motor performance.*

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SOLID PROPELLANTS

- Solid propellants are classified into the following four categories:
  - (*a*) Double-base or homogeneous propellants,
  - (*b*) Composite or heterogeneous propellants,
  - (*c*) Composite modified double-base propellants, and
  - (*d*) Nitramine propellants.
    - **high-energy solid propellant** in which a significant portion of the energetic material is a **nitramine compound**. $-N-NO_2$

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SOLID PROPELLANTS

- Solid propellants are classified into the following four categories:
- (*a*) Double-base or homogeneous propellants
  - A **double-base propellant** mainly contains two energetic ingredients:
    - **Nitrocellulose (NC)** → fuel-rich component
    - **Nitroglycerin (NG)** → energetic plasticizer
  - **no separate solid oxidizer particle dispersed throughout the binder**, as in a composite propellant.

- **Example:** Many traditional gun and rocket propellant formulations use the double-base concept.

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SOLID PROPELLANTS

- **Composite or heterogeneous propellants**
- A **composite propellant** consists of physically different ingredients that remain as separate phases.
- For example, a common conceptual formulation contains:
  - **Ammonium perchlorate (AP)** → oxidizer
  - **Polymer binder** → fuel/binder
  - **Aluminium** → additional fuel, in some formulations
  - Small quantities of other additives
  - The oxidizer exists as **solid particles distributed throughout the binder**.

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SOLID PROPELLANTS

- **Composite Modified Double-Base (CMDB) propellants**
  - CMDB propellants combine features of **double-base** and **composite** propellants.
- Double-base matrix + solid oxidizer particles

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SOLID PROPELLANTS

- Common nitramines include:
  - **RDX** – Research Department Explosive
  - **HMX** – High Melting Explosive
  - **CL-20** – a more energetic nitramine, used in specialized formulations

- contain both **nitrogen and oxygen-rich energetic groups**, allowing them to decompose rapidly and release large amounts of hot gaseous products.

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Solid propellant combustion and Internal Ballistics of Motors

- A rocket motor's **operation and its design** depend on the propellant's combustion characteristics such as
  - burning rate,
  - burning surface, and
  - grain geometry

- The branch of applied science describing the above is known as ***internal ballistics.***

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Solid propellant combustion and Internal Ballistics of Motors

- The burning surface of a propellant grain recedes in a direction essentially perpendicular to it.

- The **rate of regression**, usually expressed in cm/sec, mm/sec, or in./sec, is the *burning rate r*

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Solid propellant combustion and Internal Ballistics of Motors

- **Burning Rate:** The speed at which the solid propellant chemical reaction consumes the material, usually measured in inches per second or millimeters per second.

- **Regression Rate:** The speed at which the solid propellant's burning surface recedes or moves backward perpendicularly into the solid grain as it turns into gas.

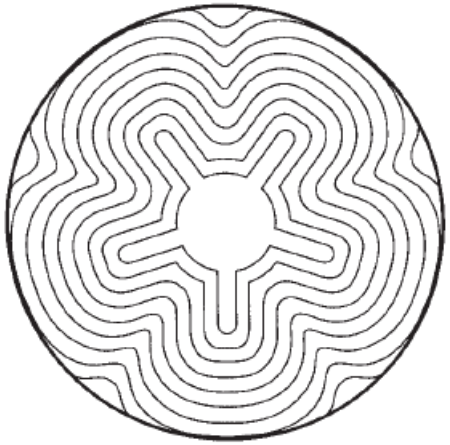
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Solid propellant combustion and Internal Ballistics of Motors

- Two-dimensional grain with five slots residing in a central cylindrical cavity.

- Rocket motor **design and development** depends significantly on
  - knowledge of burning rate behavior of the selected propellant under all motor operating conditions and design limits.

- **Diagram of successive burning surface contours, each a fixed small time interval apart.**

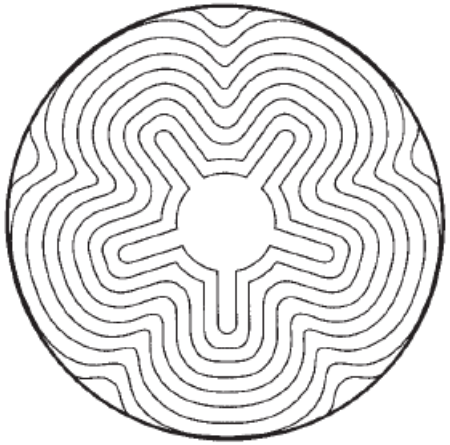


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Solid propellant combustion and Internal Ballistics of Motors

- When the **lengths of these contour lines** are roughly the same (within ±15%), the burning area is considered to be constant.
- The burning surface area A_b diminishes greatly near the end causing reduced thrust and chamber pressure.
- When the **pressure falls below the conflagration value**, combustion will stop leaving some unburnt propellant (slivers).

- **Diagram of successive burning surface contours, each a fixed small time interval apart.**



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Solid propellant combustion and Internal Ballistics of Motors

- **For any given propellant formulation** the burning rate can be increased or otherwise modified by the following:
  - 1. Higher temperatures of solid propellant prior to start
  - 2. Higher combustion gas chamber pressures
  - 3. Higher combustion gas temperatures
  - 4. Higher gas flow velocities parallel to its burning surface
  - 5. Rocket motor motions (acceleration and spin-induced grain stress)

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Solid propellant combustion and Internal Ballistics of Motors

- **Burning rate** is a function of the propellant composition.

- For composite propellants it can be increased by changing the propellant characteristics as follows:
  - 1. Add a burning rate ***catalyst***, often called burning rate *modifier* (0.1 to 3.0% of propellant) or increase percentage of existing catalyst.
    - small quantity of material added to the propellant to **change the rate at which the propellant burns**.
    - For increasing burning rate, the modifier acts mainly by **accelerating the chemical reactions in the combustion zone**.

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Solid propellant combustion and Internal Ballistics of Motors

– 2. *Decrease* the *oxidizer particle size*.

- For the **same mass of oxidizer**, smaller particles provide a much larger total surface area.
- Therefore, **mixing and reaction** between the oxidizer and fuel occur more rapidly, increasing the burning rate.

– 3. *Increase oxidizer percentage*.

- depends on the particular fuel, oxidizer, binder and formulation.

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Solid propellant combustion and Internal Ballistics of Motors

- 4. Increase the *heat of combustion* of the binder and/or the plasticizer.
  - The **binder** has two roles:
    - It holds the propellant ingredients together.
    - It also contributes fuel to the combustion process.
    - A **plasticizer** is added mainly to modify the mechanical and processing properties of the binder, and it can also contribute energy.

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Solid propellant combustion and Internal Ballistics of Motors

– 5. Imbed *wires* or *metal staples* in the propellant.

- The metal conducts heat into the surrounding propellant. This preheats the material ahead of the burning surface.

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- A propellant grain has a web thickness of **40 mm** and burns at a constant rate of **8 mm/s**.
- Determine the approximate burning time.

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- A propellant grain has a web thickness of **40 mm** and burns at a constant rate of **8 mm/s**.
- Determine the approximate burning time.

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$$t_b = \frac{w}{r_b}$$

$$t_b = \frac{40}{8}$$

$$\boxed{t_b = 5\ s}$$

- **A single-stage rocket requires:**

$$\Delta V = 4500\, m/s$$

- **and has effective exhaust velocity:**

$$V_e = 3000\, m/s$$

- **Find the required initial-to-final mass ratio.**

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A single-stage rocket requires: $\Delta V = 4500\ m/s$
and has effective exhaust velocity: $V_e = 3000\ m/s$
Find the required initial-to-final mass ratio.

$$\Delta V = V_e \ln\left(\frac{m_0}{m_f}\right)$$

$$\frac{m_0}{m_f} = e^{\Delta V / V_e}$$

$$\frac{m_0}{m_f} = e^{4500/3000}$$

$$= e^{1.5}$$

$$\boxed{\frac{m_0}{m_f} = 4.48}$$

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Grain designs have to satisfy several interrelated *requirements*:

- 1. ***Rocket motor requirements*** can be determined from the ***flight mission***. They are usually established by the vehicle designers. Motor requirements can include
  - total impulse,
  - desired thrust–time curve and tolerance thereon,
  - Motor mass,
  - ambient temperature limits during storage and operation,
  - available vehicle volume or envelope,
  - desired location or movement of rocket motor's center of gravity, and
  - accelerations caused by vehicle forces (vibration, bending, aerodynamic loads, etc.).

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Grain designs have to satisfy several interrelated *requirements*:

- 2. The ***grain geometry*** is selected to fit motor requirements; it should be compact efficiently using the available volume, have an
  - appropriate burn surface versus time profile to match the desired thrust–time curve, and
  - avoid or predictably control possible erosive burning (many motors with progressive burning can tolerate short periods of erosive burning).

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Grain designs have to satisfy several interrelated *requirements*:

- Any remaining unburned propellant slivers, and often also the shift of the center of gravity during burning, should be minimized.

- Selection of the geometry can be complex.

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Grain designs have to satisfy several interrelated *requirements*:

- 3. **Grain *structural integrity***, including its liner and/or insulator, must be analyzed to assure that the grain will not fail from stress or strain under all conditions of
  - loading,
  - acceleration, or
  - thermal stress.

- Grain geometry can be changed to reduce excessive stresses.

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Grain designs have to satisfy several interrelated *requirements*:

- 4. Any ***internal cavity volume*** made of perforations, slots, ports, and fins increases with burning time. These cavities need to be evaluated for
  - resonance,
  - vibration damping, and
  - *combustion stability*.

- 5. The ***processing*** **of the grain and the** ***fabrication*** of the propellant should be repeatable, simple, low cost and only cause acceptable thermal stresses.

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$$r_b = a p_c^n$$

$$\dot{m} = \rho_p A_b r$$

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PROBLEM

- An end-burning rocket uses a cylindrical double-base propellant grain with a diameter of 200 mm and generates a thrust of 350 N over a period of 300 s. The thrust coefficient is 1.15. The characteristics of the propellant are:
- Density of propellant grain = 1500 kg/m^3
- a = 4 mm/s; n = 0.5
- C* =1500 m/s
- Determine: (i) the length of the grain
- (ii) throat diameter of the nozzle

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GIVEN DATA

- Grain diameter, $D = 200$ mm $= 0.2$ m
- Thrust, $F = 350$ N
- Burning time, $t_b = 300$ s
- Thrust coefficient, $C_F = 1.15$
- Propellant density, $\rho_p = 1500$ kg/m^3
- Burning-rate coefficient, $a = 4$ mm/s $= 0.004$ m/s
- Pressure exponent, $n = 0.5$
- Characteristic velocity, C∗=1500 m/s

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